Data Cleaning & Transformation Documentation

Power Query Steps, Data Modeling & Star Schema Overview

Schema Architecture: `Star Schema`

This document records all data preparation, cleaning, and transformation steps applied across the dimension and fact tables to ensure data integrity and optimal analytical modeling.

1. Dimension Tables Transformation

Dim_Doctor

Promoted headers and updated data types for all columns. Name casing was preserved without modification.

Dim_Patient

Applied capitalized casing to patient names to maintain consistent formatting across all reporting views.

1 Transformation Steps: Dim_Doctor

Power Query Command	Description & Purpose
`Use First Row as Headers`	Promoted the first row of raw data to official table column headers.
`Change Type`	Validated and set proper data types for all table fields.

2 Transformation Steps: Dim_Patient

Power Query Command	Description & Purpose
`Capitalize Each Word`	Capitalized the first letter of each word in the patient name column for uniform presentation.

2. Fact Table Transformation

Fact Table Summary: Updated field data types to streamline date-based filtering, facilitate time-intelligence calculations, and ensure precision for financial calculations.

3 Data Type Adjustments - Fact Table

Column Name	Previous Type	New Type	Transformation Purpose
Date of Admission	`Date / Time`	`Date`	Removed timestamp component for cleaner date hierarchy integration.
Discharge Date	`Date / Time`	`Date`	Standardized to date format for simplified length-of-stay metrics.
Billing Amount	`Whole Number`	`Fixed Decimal Number`	Converted to currency/fixed decimal format for accurate financial aggregations.

3. Data Quality & Modeling Best Practices

Date Dimension Alignment

Converting datetime fields to pure `Date` types avoids relationship cardinality issues when joining with a standard Date table.

Financial Precision

Using `Fixed Decimal Number` prevents floating-point rounding errors during large-scale measure calculations and reporting.